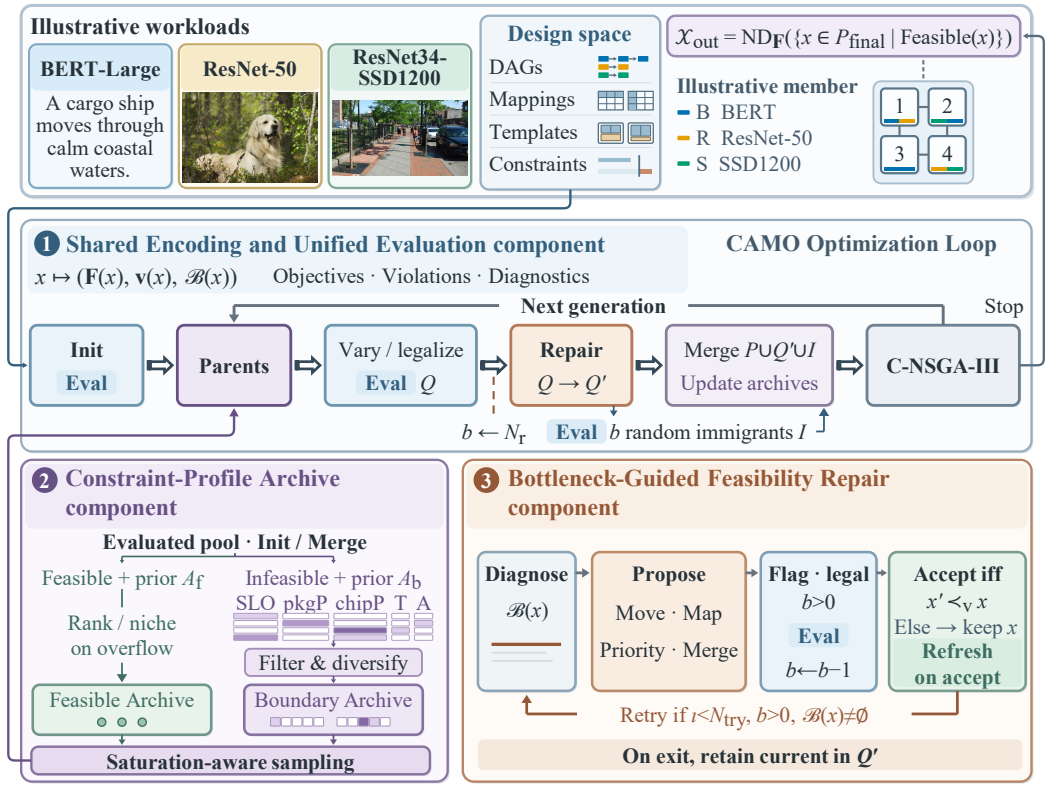

Illustrative workloads
BERT-Large
A cargo ship moves through calm coastal waters.
ResNet-50
ResNet34-SSD1200
Design space
DAGs
Mappings
Templates
Constraints
$\mathcal{X}_{\text{out}} = \text{ND}_{\mathbf{F}}(\{x \in P_{\text{final}} \mid \text{Feasible}(x)\})$
Illustrative member
B BERT
R ResNet-50
S SSD1200
1
2
3
4
1 Shared Encoding and Unified Evaluation component
$x \mapsto (\mathbf{F}(x), \mathbf{v}(x), \mathscr{B}(x))$
Objectives · Violations · Diagnostics
CAMO Optimization Loop
Next generation
Stop
Init
Eval
Parents
Vary / legalize
Eval $Q$
Repair
$Q \to Q'$
Merge $P \cup Q' \cup I$
Update archives
C-NSGA-III
$b \leftarrow N_{\text{r}}$
Eval $b$ random immigrants $I$
2 Constraint-Profile Archive component
Evaluated pool · Init / Merge
Feasible + prior $A_{\text{f}}$
Rank / niche on overflow
Infeasible + prior $A_{\text{b}}$
SLO pkgP chipP T A
Filter & diversify
Feasible Archive
Boundary Archive
Saturation-aware sampling
3 Bottleneck-Guided Feasibility Repair component
Diagnose
$\mathscr{B}(x)$
Propose
Move · Map
Priority · Merge
Flag · legal
$b>0$
Eval
$b \leftarrow b-1$
Accept iff
$x' \prec_{\text{v}} x$
Else → keep $x$
Refresh on accept
Retry if $\iota<N_{\text{try}}$, $b>0$, $\mathscr{B}(x)\neq\emptyset$
On exit, retain current in $Q'$